

Can a Novel Virtual Reality Simulator, Developed for Standalone HMDs, Effectively Prepare Patients for an MRI Examination?

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ABSTRACT

Magnetic Resonance Imaging (MRI) examinations can be scary for pediatric or claustrophobic patients. Virtual reality (VR) simulation has shown the potential to prepare patients for MRI and alleviate anxiety. However, existing VR simulations either use low-quality mobile headsets or require a complex setup, limiting immersion level and practicality. No existing simulation incorporates interactive elements to train patients to hold still. We have designed a novel VR-MRI simulator for standalone HMDs that offers high-quality and interactive MRI simulation. After comparing it with standard preparatory material, we conclude that our VR-MRI could be more engaging, satisfying, and effective for MRI preparation.

Index Terms: [J.3] Life and medical science—magnetic resonance imaging—MRI simulation—; claustrophobia—pediatric anxiety—mock MRI—virtual reality MRI

1 INTRODUCTION

Over the last decade, the use of Magnet Resonance Imaging (MRI) has become increasingly popular due to its non-invasive and radiation-free nature. However, previous reports have shown that 1 to 15% of all patients suffer from claustrophobia, resulting in over 2 million MRI procedures being terminated [1]. Among all MRI patients, pediatric patients are more sensitive to multiple MRI stress factors and the MRI termination rate of unseated children reaches 47% (aged 2-7 years) [2]. We have developed a highly implementable virtual reality MRI (VR-MRI) simulator to better prepare patients for MRI and reduce the termination rate. The goal of this study is to evaluate the effectiveness of this novel VR-MRI relative to standard preparatory material for MRI preparation, as measured by subjective user feedback.

2 METHOD

2.1 Standard Material & VR-MRI

The standard material consists of three 50-60 seconds short training videos explaining MRI preparations, procedures, and important features, including the MRI noise and emergency squeeze ball. It provides similar information to our VR-MRI but in different forms.

The VR-MRI simulator was designed and created by the Incubator for Medical Mixed and Extended Reality at Stanford University (IMMERS). This collaborative effort received consultation from the Stanford Chariot Program, the Stanford Cognitive and Systems Neuroscience Laboratory, and Stanford Healthcare. When using our VR-MRI on standalone head-mounted displays (HMDs), caregivers can monitor the user's experience on their tablet via the real-time casting feature. Our VR-MRI requires minimal set-up, and caregivers can choose to launch each section of the simulator

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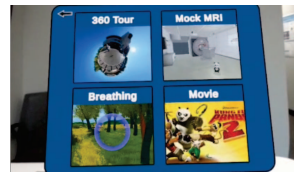


Figure 1: Home panel

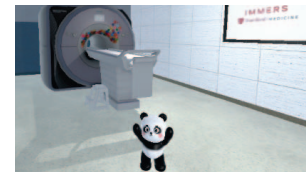


Figure 2: Panda avatar demo

to best fit patients' needs. The hold-still exercise of our simulator also serves as an evaluation tool for caregivers to predict patients' MRI success rate prior to the MRI scan. Following age selection, a section selection panel (home) appeared with four sections: 360° video tour, an MRI simulation experience with a hold-still exercise, a breathing exercise, or a 10-minute movie (Figure 1).

2.1.1 360° Video Tour

The 360° video tour section was designed to familiarize the user with the hospital environment and the necessary steps preceding the MRI scan. It introduces the pre-screen checklist, MRI squeeze ball, and medical scrubs.

2.1.2 MRI Simulation

To engage the pediatric user, a cute panda avatar started explaining how MRI worked and demonstrated the MRI procedure (Figure 2). After the tutorial, the user was prompted to lie on the real-world bed with the see-through feature of the HMD enabled. The squeeze ball feature was explained with animation. While lying on the MRI bed, a head coil was gradually placed around the virtual camera, the user's head, and the MRI bed was programmed to gradually move the user into the MRI bore. The controller was rendered as an MRI squeeze ball and squeezing it would terminate the simulation. The MRI bed stopped moving after reaching a designated location where an interactive hold-still exercise commenced, accompanied by MRI noise captured from a real MRI machine. Upon completing the exercise, the MRI bed slid out, and the home panel reappeared automatically. The adult version differentiated itself from the pediatric version mainly through changes in the narration of the panda avatar.

2.1.3 Interactive Hold-Still Exercise

A short version of this exercise was nested within the mock MRI section. A longer version was offered as one of the four sections, the 10-minute movie, in the home panel and could be launched independently. When positioned inside the MRI bore and surrounded by the head coil, the user would see a movie clip playing on a screen attached to the head coil. The green central circle marked the allowed zone where the movie would only play if the user maintained their head (indicated by a black marker) in a steady position within this zone. The yellow zone represented a buffer area; if the user's head marker was inside, the movie would pause, and the green zone would become visible (Figure 3). Clear audio and visual cues were provided to guide the user on moving their head back to the green zone. If the head marker entered the red zone, the movie would



Figure 3: Movie clip paused



Figure 4: Movie clip gone

Table 1: Mean rank in post-experience survey.

	Standard Material		VR-MRI ^a		Z	p	r
	Mean	SD	Mean	SD			
How anxious did you feel?	1.50	1.092	1.14	.363	-1	.317	.27
How engaging was this experience?	2.64	1.336	4.79	.426	-3.09	.002**	.83
How satisfying was this experience?	2.71	1.490	4.64	.633	-2.86	.004**	.76
How prepared do you feel for future MRI?	3.86	1.406	4.64	.497	-2.12	.034*	.57
I agree that I remembered to hold still.	4.14	1.231	4.86	.363	-2.06	.039*	.55
I agree that I remembered the noise.	3.79	1.528	4.79	.426	-2.14	.033*	.57

^aVR-MRI: virtual reality magnetic resonance imaging

*P < .05. **P < .01.

disappear until the user stopped looking around and refocused on the green zone (Figure 4). This hold-still exercise was programmed and designed to enhance the user's ability to remain perfectly still during an actual MRI scan. Caregivers could also use it to evaluate patients' capacity to hold still.

2.1.4 Breathing & Meditation

The breathing and meditation section served as a mindfulness-based virtual therapy to decrease the user's anxiety. The user was placed inside a cloud forest with vivid trees, grass, and sky. The user was then guided by a narrator to start the meditation with a 2-minute breathing exercise.

2.2 Study Design

This pilot study was designed to compare the adult user-reported anxiety level, engagement, and satisfaction scores between VR-MRI and a standard training video in preparing patients for MRI exams. The primary goal was to evaluate the effectiveness of the two approaches for MRI preparation, and the secondary goal was to assess user anxiety levels, engagement, and satisfaction scores for both experiences.

Participants were excluded from the study if they had mental, visual or auditory disability, and were unable to understand or communicate in English. All participants provided informed consent and underwent both preparation experiences. The order in which they experienced the two approaches was randomly determined, with participants divided into either Group 1 or Group 2. Group 1 started with the VR-MRI simulation, while Group 2 began with the standard material. Data were collected at baseline, after the first experience, after the second experience, and at the end of the study. A Likert scale from 1 (none) to 5 (extremely) was used to assess the user's subjective belief. The SPSS (version 29 by IBM) software was used for statistical analysis. Wilcoxon signed ranks test was applied to statistically analyze the differences in post-experience surveys.

3 RESULTS

A total of 14 participants were enrolled and completed the study (mean age 35.9 years). Most participants were female (8/14, 57%) and had history with virtual reality (13/14, 93%). As in Table 1, participants had similar anxiety levels after experiencing standard

Table 2: Comparison of self-reported overall preferences.

	Standard Material	VR-MRI	Same
Which experience felt more immersive? (n)	0	14	0
Which experience felt more fun? (n)	0	14	0
Which experience are you more satisfied with? (n)	0	12	2
Which experience better prepared you for future MRI? (n)	1	12	1
If you could do VR, which experience do you prefer? (n)	0	12	2

material (mean 1.50, SD 1.092) and VR-MRI (mean 1.14, SD .363) compared with baseline level (mean 1.36, SD .74). No statistically significant difference was found in participants' anxiety level ($p = .317$) post two experiences. The test revealed that participants were statistically more engaged in VR-MRI experience (mean 4.79, SD .426) than standard material (mean 2.64, SD 1.336), $z = -3.09$, $p = .002$, with a large effective size, $r = .83$. Moreover, participants were statistically more satisfied ($z = -2.86$, $p = .004$, $r = .76$) with the VR-MRI (mean 4.64, SD .633) than standard material (mean 2.71, SD 1.49). Based on participants' self-reported rating, the VR-MRI experience was statistically more effective ($z = -2.12$, $p = .034$) than the standard material in preparing them for an MRI scan. Specifically, the VR-MRI experience was statistically more effective in communicating the importance of holding still ($z = -2.06$, $p = .039$) and the element of MRI noise ($z = -2.14$, $p = .033$) than the standard material.

Participants were asked to rank their overall rating after completing both the VR-MRI and standard material (Table 2). Overall, most participants (12/14) preferred VR-MRI over standard material in preparing them for an MRI scan while one participant preferred standard material and another participant did not think it mattered.

4 CONCLUSION & DISCUSSION

The VR-MRI simulator, featuring four unique sections, shows promise as an engaging, satisfying, and effective tool for MRI preparation. Additional research is needed to validate and assess its effectiveness with real patients. With certain enhancements, virtual simulation can be used in clinical or non-clinical settings for MRI preparation or success evaluation, demonstrating significant future potential.

Our study has several limitations. We focused on evaluating self-reported survey data, which may inaccurately reflect participants' preparedness for future MRI exams. Our study results are also limited by the small sample size of volunteers. Although our VR-MRI was designed for both adult and pediatric patients, we only evaluated the adult version. In the future, a larger-scale study on real hospital patients should be conducted to test both versions of our simulator. The success rate of non-sedated MRI exams after VR-MRI simulation should be analyzed to evaluate the actual effectiveness of our simulator for MRI preparation.

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